

IKT215 Exercise 4: Classification Rules

February 05th, 2026

Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails (manuel.s.mathew@uia.no).

Question 1

Using the iris dataset, implement a classifier using **Nearest Centroid Classification Rule**. Visualize the results.

Tasks

- Exploratory Analysis (Optional here, but usually a good practice)
- Load the iris dataset and use any two features.
- Split the data for training and testing (e.g. 4:1)
- Compute class centroids (mean of features for each class).
- Using an appropriate distance metric (e.g. Euclidean distance), develop a nearest centroid classifier.
- Classify test samples by assigning them to the class of the nearest centroid.
- Visualize your results.
- Calculate an accuracy score for the classifier on the test set.

Question 2

Using the iris dataset, implement a classifier using **Nearest Neighbor Classification Rule**. Visualize the results.

Tasks

- Exploratory Analysis (Optional here, but usually a good practice)
- Load the iris dataset and use any two features.
- Split the data for training and testing (e.g. 4:1)
- Using an appropriate distance metric (e.g. Euclidean distance), develop a nearest neighbor classifier.
- Classify test samples by assigning them to the class of the nearest neighbor.
- Visualize your results.
- Calculate an accuracy score for the classifier on the test set and compare it with above.

Question 3

A medical screening test categorizes patients into 5 risk groups (1-5) based on certain parameters. You have a training dataset of 20 patients where each patient belongs to one of these risk groups and has been diagnosed as either healthy (0) or sick (1). Implement the **Discrete Histogram Classification Rule** to predict the diagnosis of new patients.

Training data is provided as follows:

Risk Group 1: 3 healthy, 1 sick

Risk Group 2: 1 healthy, 4 sick

Risk Group 3: 2 healthy, 3 sick

Risk Group 4: 4 healthy, 1 sick

Risk Group 5: 1 healthy, 0 sick